

Problem-Solution Fit

Date	14 July 2026
Team ID	N/A (Solo Project)
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

• PURPOSE • Help farmers choose the right crop for their land using data instead of guesswork, to improve yield and reduce loss.	• TRIGGER TO ACT • Start of a new sowing season, or after a poor harvest caused by an unsuitable crop choice.	• EMOTION: BEFORE • Uncertain, anxious, and reliant on guesswork or word-of-mouth advice.
• PEOPLE • Small and marginal farmers, agricultural advisory centers, and agriculture students.	• PROBLEM • Farmers often select crops based on tradition or intuition rather than actual soil nutrients (N, P, K) and climate conditions, leading to reduced yield, wasted resources, and financial loss.	• EMOTION: AFTER • Confident and reassured, with a clear, data-backed crop recommendation.
• CUSTOMER CONSTRAINTS • Limited digital literacy, unreliable internet access in rural areas, and limited access to soil testing labs.	• EXISTING ALTERNATIVES • Traditional experience-based selection, advice from other farmers, and generic government advisory pamphlets.	• HIGH-LEVEL CONCEPT • A Flask-based web application where farmers enter soil and climate values and receive an instant, Random Forest-powered crop recommendation with farming tips.